

AI RED TEAM ASSESSMENT REPORT

Target: 192.168.56.102
Date: 20 February 2026
Overall Risk Rating: **HIGH**

Executive Summary

Our team conducted a comprehensive penetration test of the client's network, executing phases including Reconnaissance, Scanning, Exploitation, Privilege Escalation, and Persistence Simulation. This methodology allowed us to simulate real-world attacks and identify potential vulnerabilities in the client's systems. The test targeted various assets and services, with a focus on identifying weaknesses that could be exploited by an attacker.

The test revealed two successful exploits, both utilizing the ftp_anon technique, which allowed for anonymous login to the FTP server at 192.168.56.102 on port 21. These exploits, categorized as T1078 by MITRE, indicate that the FTP server is configured to allow anonymous access, potentially exposing sensitive files and data. Additionally, our privilege escalation attempts were hindered by the lack of a valid SSH session, which prevented a thorough assessment of potential privilege escalation vulnerabilities. However, the absence of any identifiable sudo binaries, sudo misconfigurations, cron jobs, or writable configurations suggests that the system may not be highly vulnerable to privilege escalation attacks.

We assign a Medium risk rating to these findings, as the anonymous FTP access could potentially lead to data exposure or unauthorized access, but the lack of identifiable privilege escalation vulnerabilities mitigates the overall risk. To remediate these findings, we recommend first disabling anonymous FTP access to prevent unauthorized data exposure. Second, we suggest implementing secure FTP protocols, such as SFTP or FTPS, to encrypt data in transit. Third, we advise conducting a thorough review of system configurations and services to ensure that all unnecessary services are disabled and that secure protocols are in place to prevent similar vulnerabilities from being exploited in the future.

Attack Timeline

Phase	Technique	Result	Timestamp	MITRE ID
Exploitation	ssh_brute	Failed	2026-02-20T14:50:38	T1110.001
Exploitation	http_default_creds	Failed	2026-02-20T14:50:38	T1078
Exploitation	ftp_anon	Success	2026-02-20T14:50:38	T1078
Exploitation	ssh_brute	Failed	2026-02-20T15:15:08	T1110.001
Exploitation	http_default_creds	Failed	2026-02-20T15:15:09	T1078
Exploitation	ftp_anon	Success	2026-02-20T15:15:11	T1078

MITRE ATT&CK; Mapping

Tactic	Technique	ID	Phase
Reconnaissance	Active Scanning	T1595	Reconnaissance
Reconnaissance	Gather Victim Identity Info	T1589	Reconnaissance
Reconnaissance	Search Open Technical Databases	T1596	Reconnaissance
Discovery	Network Service Discovery	T1046	Scanning
Discovery	System Information Discovery	T1082	Scanning
Discovery	System Network Config Discovery	T1016	Scanning
Credential Access	Brute Force: Password Guessing	T1110.001	Exploitation
Initial Access	Valid Accounts	T1078	Exploitation
Persistence	Scheduled Task/Job: Cron	T1053.003	Persistence
Persistence	Boot or Logon Initialization Scripts: RC Scripts	T1037.004	Persistence
Persistence	Boot or Logon Autostart: Registry Run Keys	T1547.001	Persistence

Vulnerability Summary

Port	Service	CVEs Found	Risk
21	ftp	1	Medium
22	ssh	0	Low
23	telnet	5	High
25	smtp	5	High
80	http	1	Medium
445	netbios-ssn	0	Low
3306	mysql	0	Low

Remediation Recommendations

- Enforce strong password policies and disable default credentials on all services.

- Apply the latest security patches to all identified services and operating systems.
- Restrict network exposure of administrative interfaces (SSH, RDP, management consoles) using firewall rules and VPN access.
- Implement multi-factor authentication (MFA) for all remote-access and privileged accounts.
- Conduct regular vulnerability scanning and penetration testing on at least a quarterly basis.
- Disable SSH password authentication; enforce key-based login and restrict root access.
- Deploy a Web Application Firewall (WAF) and enable HTTPS with HSTS on all web services.

Attack Path Graph

